

LAB 3: EXPLORING UDP WITH DNS USING WIRESHARK

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SECTION: C

PART 1: DNS ANALYSIS

1. Observe DNS Caching:

```
>> dig google.com

; <<>> DiG 9.20.17 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41434
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 199     IN      A      142.250.205.206

;; Query time: 7 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:51:35 IST 2026
;; MSG SIZE  rcvd: 55
```

```

> dig google.com

; <<>> DiG 9.20.17 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50874
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                192     IN      A      142.250.205.206

;; Query time: 20 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:51:43 IST 2026
;; MSG SIZE rcvd: 55

```

No.	Time	Source	Destination	Protocol	Length	Info
94	7.545681	10.1.20.103	192.168.3.2	DNS	93	Standard query 0xa1da A google.com OPT
95	7.552904	192.168.3.2	10.1.20.103	DNS	97	Standard query response 0xa1da A google.com A 142.250.205.206 OPT
224	15.04388	10.1.20.103	192.168.3.2	DNS	93	Standard query 0xc6ba A google.com OPT
225	15.06382	192.168.3.2	10.1.20.103	DNS	97	Standard query response 0xc6ba A google.com A 142.250.205.206 OPT
326	20.82647	10.1.20.103	192.168.3.2	DNS	78	Standard query 0x9893 A ping.archlinux.org
327	20.82654	10.1.20.103	192.168.3.2	DNS	78	Standard query 0x618c AAAA ping.archlinux.org
328	20.83234	192.168.3.2	10.1.20.103	DNS	117	Standard query response 0x9893 A ping.archlinux.org CNAME redirect.archlinux.org A 95.216.1
329	20.83234	192.168.3.2	10.1.20.103	DNS	129	Standard query response 0x618c AAAA ping.archlinux.org CNAME redirect.archlinux.org AAAA 2a


```

Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
- Queries
  - google.com: type A, class IN
    Name: google.com
    [Name Length: 10]
    [Label Count: 2]
    Type: A (1) (Host Address)
    Class: IN (0x0001)
- Additional records
  - <Root>: type OPT
    [Options: 0x0001]

```

2. Query for Other Record Types:

try `dig www.google.com AAAA` to observe the IPv6 address.

```
> dig google.com AAAA

; <<>> DiG 9.20.17 <<>> google.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 2026
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                IN      AAAA

;; ANSWER SECTION:
google.com.                205     IN      AAAA    2404:6800:4007:83c::200e

;; Query time: 9 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:29:11 IST 2026
;; MSG SIZE rcvd: 67
```

udp.port==53						
No.	Time	Source	Destination	Protocol	Length	Info
70	3.818451	10.1.20.103	192.168.3.2	DNS	93	Standard query 0x07ea AAAA google.com OPT
71	3.827056	192.168.3.2	10.1.20.103	DNS	109	Standard query response 0x07ea AAAA google.com AAAA 2404:6800:4007:83c::200e OPT
197	9.667634	10.1.20.103	192.168.3.2	DNS	78	Standard query 0x37ba A ping.archlinux.org
198	9.667699	10.1.20.103	192.168.3.2	DNS	78	Standard query 0x39c2 AAAA ping.archlinux.org
199	9.690133	192.168.3.2	10.1.20.103	DNS	117	Standard query response 0x37ba A ping.archlinux.org CNAME redirect.archlinux.org A 95.216.1
200	9.690135	192.168.3.2	10.1.20.103	DNS	129	Standard query response 0x39c2 AAAA ping.archlinux.org CNAME redirect.archlinux.org AAAA 2a

Transaction ID: 0x07ea
Flags: 0x0120 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
- google.com: type AAAA, class IN
Name: google.com
[Name Length: 10]
[Label Count: 2]
Type: AAAA (28) (IPv6 Address)
Class: IN (0x0001)
- Additional records
<Root>: type OPT

Use `dig google.com MX` to view mail server records.

udp.port==53						
No.	Time	Source	Destination	Protocol	Length	Info
40	1.728287...	10.1.20.103	192.168.3.2	DNS	93	Standard query 0x8ef5 MX google.com OPT
41	1.733652...	192.168.3.2	10.1.20.103	DNS	182	Standard query response 0x8ef5 MX google.com MX 10 smtp.google.com A 142.251.10.27 A 142...

Frame 40: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface wlan0, id 0
Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)
Internet Protocol Version 4, Src: 10.1.20.103, Dst: 192.168.3.2
User Datagram Protocol, Src Port: 51804, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x8ef5
Flags: 0x0120 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
google.com: type MX, class IN
Name: google.com
[Name Length: 10]

```
▶ dig google.com MX
```

```
; <<>> DiG 9.20.17 <<>> google.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 44515
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 6

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;google.com.                IN      MX

;; ANSWER SECTION:
google.com.                205     IN      MX      10 smtp.google.com.

;; ADDITIONAL SECTION:
smtp.google.com.          148     IN      A        142.251.12.26
smtp.google.com.          148     IN      A        142.251.12.27
smtp.google.com.          148     IN      A        142.251.10.26
smtp.google.com.          148     IN      A        64.233.170.27
smtp.google.com.          148     IN      A        142.251.10.27

;; Query time: 6 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (UDP)
;; WHEN: Thu Feb 12 14:46:33 IST 2026
;; MSG SIZE rcvd: 140
```

3. Modify DNS Servers:

```
>> dig @1.1.1.1 www.google.com

; <<>> DiG 9.20.17 <<>> @1.1.1.1 www.google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49018
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 1232
;; QUESTION SECTION:
;www.google.com.                IN      A

;; ANSWER SECTION:
www.google.com.                42      IN      A      142.251.221.228

;; Query time: 58 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Thu Feb 12 14:58:17 IST 2026
;; MSG SIZE rcvd: 59
```

```

>> dig @8.8.8.8 www.google.com

; <<>> DiG 9.20.17 <<>> @8.8.8.8 www.google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3933
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;; udp: 512
;; QUESTION SECTION:
;www.google.com.                IN      A

;; ANSWER SECTION:
www.google.com.                179     IN      A      142.251.221.164

;; Query time: 17 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Thu Feb 12 14:58:29 IST 2026
;; MSG SIZE rcvd: 59

```

› Cloudflare: 58ms

› Google: 17ms

udp.port==53						
No.	Time	Source	Destination	Protocol	Length	Info
141	4.704161...	10.1.20.103	1.1.1.1	DNS	97	Standard query 0xbf7a A www.google.com OPT
150	4.762275...	1.1.1.1	10.1.20.103	DNS	101	Standard query response 0xbf7a A www.google.com A 142.251.221.228 OPT
474	16.23039...	10.1.20.103	8.8.8.8	DNS	97	Standard query 0xf5d A www.google.com OPT
475	16.24771...	8.8.8.8	10.1.20.103	DNS	101	Standard query response 0xf5d A www.google.com A 142.251.221.164 OPT

Questions: 1
 Answer RRs: 0
 Authority RRs: 0
 Additional RRs: 1

Queries

- www.google.com: type A, class IN
 - Name: www.google.com
 - [Name Length: 14]
 - [Label Count: 3]
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)
- Additional records
 - <Root>: type OPT

[Response In: 150]

4. Analyze DNS over TCP:

```
>> dig +tcp www.google.com

; <<>> DiG 9.20.17 <<>> +tcp www.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 57961
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;www.google.com.                IN      A

;; ANSWER SECTION:
www.google.com.                100     IN      A      142.251.220.4

;; Query time: 4 msec
;; SERVER: 192.168.3.2#53(192.168.3.2) (TCP)
;; WHEN: Thu Feb 12 15:10:23 IST 2026
;; MSG SIZE rcvd: 59
```

No.	Time	Source	Destination	Protocol	Length	Info
67	1.969555...	10.1.20.103	192.168.3.2	TCP	74	45723 → 53 [SYN] Seq=0 Win=64660 Len=0 MSS=1220 SACK_PERM TSval=3505431216 TSecr=0 WS=1024
68	1.974328...	192.168.3.2	10.1.20.103	TCP	74	53 → 45723 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM TSval=276366777 TSecr=276366777
69	1.974386...	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=3505431221 TSecr=276366777
70	1.974509...	10.1.20.103	192.168.3.2	DNS	123	Standard query 0xe269 A www.google.com OPT
71	1.979779...	192.168.3.2	10.1.20.103	DNS	127	Standard query response 0xe269 A www.google.com A 142.251.220.4 OPT
72	1.979831...	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [ACK] Seq=58 Ack=62 Win=65536 Len=0 TSval=3505431226 TSecr=276366777
73	1.980674...	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [FIN, ACK] Seq=58 Ack=62 Win=65536 Len=0 TSval=3505431227 TSecr=276366777
74	1.983936...	192.168.3.2	10.1.20.103	TCP	66	53 → 45723 [ACK] Seq=62 Ack=59 Win=2098176 Len=0 TSval=276366786 TSecr=3505431227
75	1.983936...	192.168.3.2	10.1.20.103	TCP	66	53 → 45723 [FIN, ACK] Seq=62 Ack=59 Win=2098176 Len=0 TSval=276366786 TSecr=3505431227
76	1.984006...	10.1.20.103	192.168.3.2	TCP	66	45723 → 53 [ACK] Seq=59 Ack=63 Win=65536 Len=0 TSval=3505431230 TSecr=276366786

Frame 67: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface wlan0, id 0

Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)

Internet Protocol Version 4, Src: 10.1.20.103, Dst: 192.168.3.2

Transmission Control Protocol, Src Port: 45723, Dst Port: 53, Seq: 0, Len: 0

Source Port: 45723

Destination Port: 53

[Stream index: 0]

[Stream Packet Number: 1]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 1252856957

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment Number (raw): 0

1010 = Header Length: 40 bytes (10)

Flags: 0x002 (SYN)

Window: 64660

[Calculated window size: 64660]

Checksum: 0xe240 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale

[Timestamps]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

PART 2: UDP COMMUNICATION

Set up a UDP server using netcat on port 54321.

```
> nc -u -l 54321
message1
^C
> nc -u -l 54321
message2
^C
> nc -u -l 54321
message3
^C
```

```
> echo "message1" | nc -u 127.0.0.1 54321
^C
> echo "message2" | nc -u 127.0.0.1 54321
^C
> echo "message3" | nc -u 127.0.0.1 54321
```

udp.port == 54321						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000...	127.0.0.1	127.0.0.1	UDP	51	40605 → 54321 Len=9
2	20.45845...	127.0.0.1	127.0.0.1	UDP	51	42520 → 54321 Len=9
3	42.43357...	127.0.0.1	127.0.0.1	UDP	51	41386 → 54321 Len=9

Frame 1: Packet, 51 bytes on wire (408 bits), 51 bytes captured (408 bits) on interface lo, id 0

Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

User Datagram Protocol, Src Port: 40605, Dst Port: 54321

- Source Port: 40605
- Destination Port: 54321
- Length: 17
- Checksum: 0xfe24 [unverified]
- [Checksum Status: Unverified]
- [Stream index: 0]
- [Stream Packet Number: 1]
- [Timestamps]
- UDP payload (9 bytes)

Data (9 bytes)

Data: 6d657373616765310a

[Length: 9]

| Write a report:

› How does UDP differ from TCP?

UDP is connectionless and does not guarantee delivery, ordering, or error recovery. It has a fixed 8 byte header and no handshake. TCP is connection-oriented, reliable, ensures ordered delivery, and uses flow and congestion control, but with higher overhead.

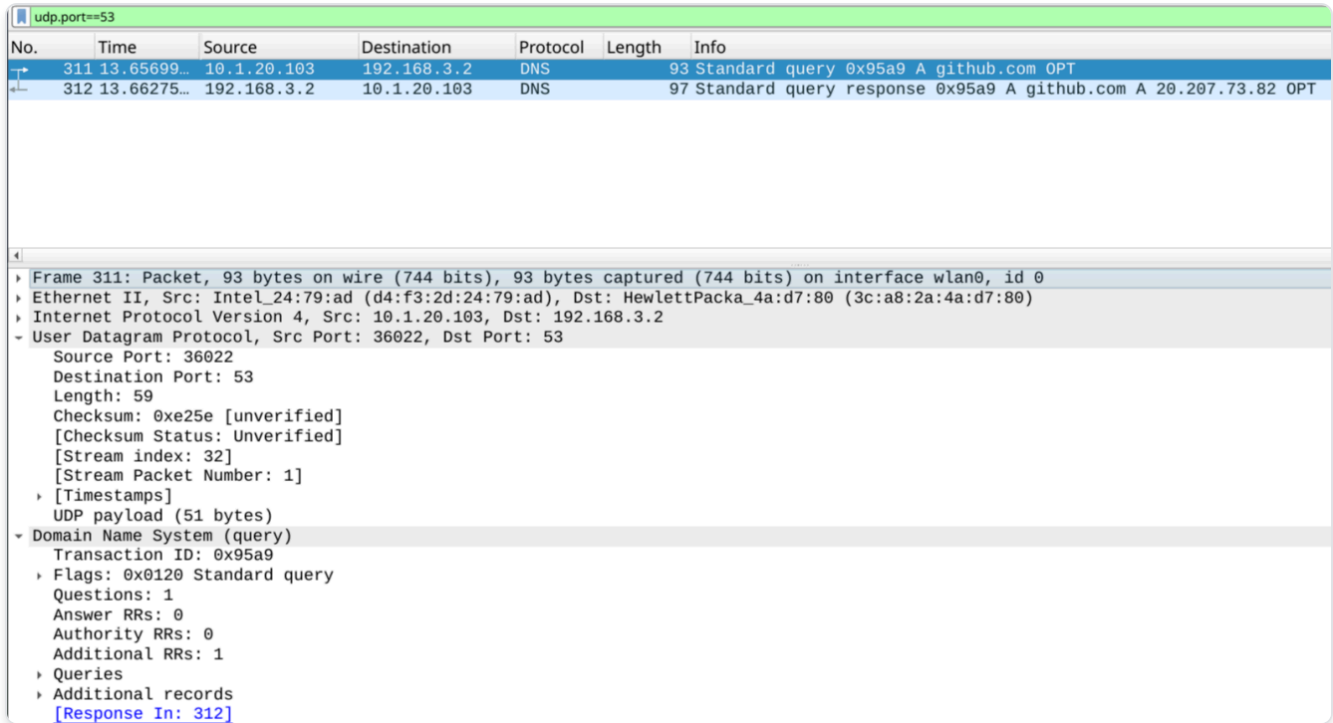
› Why is UDP suitable for DNS?

DNS queries are small and require fast responses. UDP avoids connection setup delays, reduces overhead, and allows DNS servers to handle large volumes of short request-response transactions efficiently. TCP is only used when responses are large or for zone transfers.

› Observations about UDP traffic in Wireshark:

UDP packets show only four header fields (Source Port, Destination Port, Length, Checksum). No handshake packets are seen. In DNS traffic, the client uses an ephemeral source port and sends to port 53, and the server reply reverses the port numbers.

PART 3: ANALYSIS OF THE UDP PROTOCOL IN DNS USING WIRESHARK



No.	Time	Source	Destination	Protocol	Length	Info
311	13.65699...	10.1.20.103	192.168.3.2	DNS	93	Standard query 0x95a9 A github.com OPT
312	13.66275...	192.168.3.2	10.1.20.103	DNS	97	Standard query response 0x95a9 A github.com A 20.207.73.82 OPT

Frame 311: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface wlan0, id 0
Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: HewlettPacka_4a:d7:80 (3c:a8:2a:4a:d7:80)
Internet Protocol Version 4, Src: 10.1.20.103, Dst: 192.168.3.2
User Datagram Protocol, Src Port: 36022, Dst Port: 53
Source Port: 36022
Destination Port: 53
Length: 59
Checksum: 0xe25e [unverified]
[Checksum Status: Unverified]
[Stream index: 32]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (51 bytes)
Domain Name System (query)
Transaction ID: 0x95a9
Flags: 0x0120 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
Additional records
[Response In: 312]

1. Select the first UDP segment in your trace

› 1. What is the packet number of this segment in the trace file?

• PACKET NUMBER: 311

› 2. What type of application-layer payload or protocol message is being carried in this UDP segment? Look at the details of this packet in Wireshark.

• PROTOCOL CARRIED: DNS STANDARD QUERY

› 3. How many fields are there in the UDP header? What are the names of these fields?

› UDP header fields: 4

Source port

Destination port

Length

Checksum

2. By consulting the displayed information in Wireshark's packet content field for this packet, what is the length (in bytes) of the UDP header field?

› UDP length (59) - UDP payload(51) -> 8bytes.

3. Examine the pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.231.151.62	10.231.151.160	DNS	93	Standard query 0x4c8f A google.com OPT
2	0.218369575	10.231.151.160	10.231.151.62	DNS	86	Standard query response 0x4c8f A google.com A 142.250.77.142
3	0.320067473	fe80::6493:acff:fe7... ff02::1:ff8c:f74		ICMPv6	86	Neighbor Solicitation for 2401:4900:901e:dcbb:9b3:66a1:e98c:f74 from 66:93:ac:73:21:fc

Frame 1: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface wlan0, id 0	0000	66 93 ac 73 21 fc d4 f3 2d 24 79 ad 08 00
Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:24:79:ad), Dst: 66:93:ac:73:21:fc (66:93:ac:73:21:fc)	0010	00 4f d4 1d 00 00 40 11 61 d4 0a e7 97 3e
Internet Protocol Version 4, Src: 10.231.151.62, Dst: 10.231.151.160	0020	97 a0 d0 41 00 35 00 3b 44 f9 4c 8f 01 20
User Datagram Protocol, Src Port: 53313, Dst Port: 53	0030	00 00 00 00 00 01 06 67 6f 6f 67 6c 65 03
Domain Name System (query)	0040	6d 00 00 01 00 01 00 00 29 04 d0 00 00 00
	0050	0c 00 0a 00 08 ae 57 00 a9 b8 c7 e8 45

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.231.151.62	10.231.151.160	DNS	93	Standard query 0x4c8f A google.com OPT
2	0.218369575	10.231.151.160	10.231.151.62	DNS	86	Standard query response 0x4c8f A google.com A 142.250.77.142
3	0.320067473	fe80::6493:acff:fe7... ff02::1:ff8c:f74		ICMPv6	86	Neighbor Solicitation for 2401:4900:901e:dcbb:9b3:66a1:e98c:f74 from 66:93:ac:73:21:fc

Frame 2: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface wlan0, id 0	0000	d4 f3 2d 24 79 ad 66 93 ac 73 21 fc 08
Ethernet II, Src: 66:93:ac:73:21:fc (66:93:ac:73:21:fc), Dst: Intel_24:79:ad (d4:f3:2d:24:79:ad)	0010	00 48 bf 4f 40 00 40 11 36 a9 0a e7 97
Internet Protocol Version 4, Src: 10.231.151.160, Dst: 10.231.151.62	0020	97 3e 00 35 d0 41 00 34 6c a9 4c 8f 81
User Datagram Protocol, Src Port: 53, Dst Port: 53313	0030	00 01 00 00 00 00 06 67 6f 6f 67 6c 65
Domain Name System (response)	0040	6d 00 00 01 00 01 c0 0c 00 01 00 01 00
	0050	00 04 8e fa 4d 8e

› A. What is the packet number of these two UDP segments in the trace file?

› 1 and 2

› B. What is the value in the source port field in this UDP segment?

- 53313

› C. What is the value in the destination port field in this UDP segment?

- 53

› D. What is the value in the source port field in this second UDP segment?

- 53

› E. What is the value in the destination port field in this second UDP segment? Describe the relationship between the port numbers in the two packets

- 53313

- THE CLIENT PICKS A TEMPORARY PORT, AND THE SERVER REPLIES BACK TO THAT SAME PORT.